

■ COMPLETE NOTES — TOPIC 02

CSS Box Model

Margin, Padding & Borders — Poora Samjho

Web Development Course — Shauray

Box Model Kya Hai

Content

Padding

Border

Margin

Shorthand Properties

box-sizing

Border Styles

Border Radius

Margin Collapse

Outline vs Border

// TOPIC 01

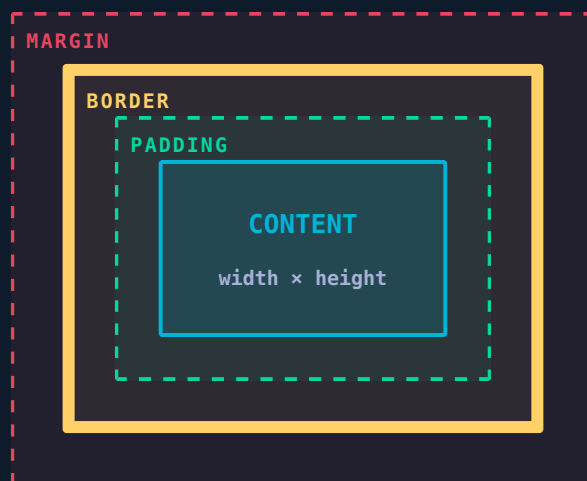
CSS Box Model Kya Hai?

Har HTML element ek box hai — yeh box 4 layers se bana hai

KYA HAI?

CSS Box Model ek **concept** hai jisme har HTML element ko ek rectangular box ki tarah treat kiya jaata hai. Is box mein 4 layers hoti hain —

Content, Padding, Border, aur Margin. Yeh samajhna CSS layout ka foundation hai — bina iske spacing aur sizing kabhi sahi nahi hogi.



Layer	Kya Hai	DevTools Color
Content	Actual text ya image — width aur height yahan apply hoti hai	Blue
Padding	Content aur border ke beech ka space — background show karta hai	Green
Border	Padding ke bahar ki line — color, style, width set hoti hai	Yellow

Layer	Kya Hai	DevTools Color
Margin	Border ke bahar ka space — elements ke beech gap — transparent hota hai	Orange

■ **Tip:** Chrome DevTools mein kisi bhi element pe right-click → Inspect karo — neeche Box Model diagram dikhega jisme exact padding, border, margin values hongii!

■ Total Element Size Kaise Calculate Hoti Hai?

box-model-size.css

```
/* Default (content-box) mein Total Size: */
/* Total Width  = width + padding-L + padding-R + border-L + border-R */
/* Total Height = height + padding-T + padding-B + border-T + border-B */

.box {
  width:    200px;
  padding:  20px;
  border:   5px solid blue;
  margin:   10px;
}

/*
  Content   = 200px
  Padding   = 20 + 20 = 40px
  Border    = 5  + 5  = 10px
  _____

  Screen pe dikhi width = 250px ← expected 200 tha!
  Margin alag — surrounding space hai, total mein count nahi
  */
```

■ **Yaad Rakho:** Default mein `width: 200px` sirf content ka size hai — puri box ka nahi! Isliye `box-sizing: border-box` use karte hain — aage detail mein aayega.

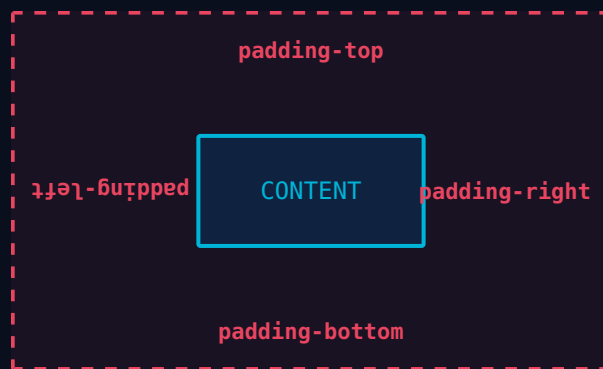
// TOPIC 02

Padding

Content aur border ke beech ka andar ka space

KYA HAI?

Padding element ke content aur uski border ke beech ka **andar ka space** hai. Padding ka background color element ke background ke saath aata hai — transparent nahi hota. Negative padding **allowed nahi** hai!



padding.css

```
/* Alag alag sides */
.box {
  padding-top:    20px;
  padding-right:  30px;
  padding-bottom: 20px;
  padding-left:   30px;
}

/* — Shorthand — */

/* 1 value — charon taraf same */
.box { padding: 20px; }

/* top=20 right=20 bottom=20 left=20 */

/* 2 values — top&bottom left&right */
```

```
.box { padding: 10px 20px; }  
  
/* top=10, bottom=10 | left=20, right=20 */  
  
/* 3 values – top left&right bottom */  
.box { padding: 10px 20px 30px; }  
/* top=10 | left=20, right=20 | bottom=30 */  
  
/* 4 values – Clockwise: TOP RIGHT BOTTOM LEFT */  
.box { padding: 10px 20px 30px 40px; }  
/* top=10, right=20, bottom=30, left=40 */
```

■ Yaad Kaise Rakhen? TRouBLE

4 values ka order hamesha
Clockwise hota hai:

T → R → B → L

Top → Right → Bottom → Left

Trick: "**TRouBLE**"

■ % mein Padding

% mein padding dene par yeh
parent ki WIDTH ke relative hoti
hai — height ke nahi!

padding: 5%;
= parent width ka 5%

Responsive square banane ke liye
useful hai!

■ **Note:** Padding **negative nahi** ho sakti — negative value dene par
browser zero maan leta hai. Negative spacing chahiye toh margin use karo!

// TOPIC 03

Margin

Elements ke beech bahar ka space — hamesha transparent

KYA HAI?

Margin element ke **bahar** ka space hai — border ke baad. Yeh hamesha **transparent** hota hai (background show nahi karta). Ek element aur doosre element ke beech distance banata hai. Margin **negative bhi ho sakta hai!**

margin.css

```
/* Alag alag sides */
.box {
    margin-top:    20px;
    margin-right:  30px;
    margin-bottom: 20px;
    margin-left:   30px;
}

/* Shorthand — padding jaisa hi */
.box { margin: 20px; }
.box { margin: 10px 20px; }
.box { margin: 10px 20px 30px 40px; }

/* — SPECIAL VALUES — */

/* auto — element horizontally CENTER ho jaata hai! */
.container {
    width: 800px;
    margin: 0 auto; /* ← center trick! */
}
```

```

/* Negative margin – overlap effect */
.overlap {
    margin-top: -20px; /* element upar khich jaata hai */
}

/* body ka default margin hatao */
body {
    margin: 0; /* browser 8px default margin remove karo */
}

```

■ Margin Collapse — Important Concept!

MARGIN COLLAPSE KYA HAI?

Jab do **block elements vertically** adjacent hote hain, unki margins **collapse** ho jaati hain — dono margins add nahi hoti, sirf **badi wali margin** apply hoti hai.

margin-collapse.css

```

.box1 { margin-bottom: 30px; }
.box2 { margin-top: 20px; }

/*
    Expect karte: 30 + 20 = 50px gap
    Actually hota: max(30, 20) = 30px ← MARGIN COLLAPSE!

    ■ Horizontal margins collapse NAHI hote
    ■ Flexbox / Grid children mein collapse NAHI hota
*/

/* Collapse rokne ke tarike: */
.parent {
    overflow: hidden; /* option 1 */
    padding-top: 1px; /* option 2 */
}

```



```
border-top: 1px solid transparent; /* option 3 */  
}
```

Property	Padding	Margin
Location	Element ke andar	Element ke bahar
Background	Background color show karta hai	Hamesha transparent
Negative Value	✗ Allowed nahi	✓ Ho sakta hai
auto Value	✗ Kaam nahi karta	✓ Centering ke liye
Collapse	✗ Nahi hoti	✓ Vertical mein hoti hai

■ **Tip:** Element ko horizontally center karne ka classic tarika: pehle `width` set karo, phir `margin: 0 auto` likho. Sabse common centering trick hai!

Border

Padding ke bahar aur margin ke andar ki visible line

KYA HAI?

Border padding aur margin ke beech ki **visible line** hai. Border ki teen properties hoti hain — **width** (kitni moti), **style** (kaisi dikhegi), aur **color** (kaunsa rang). Teeno dena zaroori hai — ek bhi miss karo toh border nahi dikhegi!

 border.css

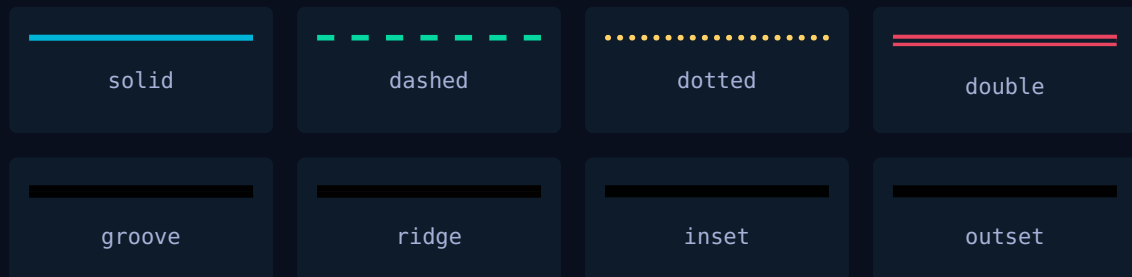
```
/* Alag alag properties */
.box {
    border-width: 2px;
    border-style: solid;
    border-color: #e94560;
}

/* Shorthand – width style color (yeh order follow karo!) */
.box {
    border: 2px solid #e94560;    /* Recommended */
}

/* Specific sides ke liye */
.box {
    border-top:    3px solid red;
    border-right:  2px dashed blue;
    border-bottom: 1px dotted green;
    border-left:   5px double purple;
}
```

```
/* Border hatana */  
.no-border { border: none; }
```

■ Border Styles — Saare Options



■ Border Radius — Rounded Corners

border-radius.css

```
.box { border-radius: 8px; } /* sab corners */  
.circle { border-radius: 50%; } /* perfect circle (w=h zaroori) */  
.pill { border-radius: 999px; } /* capsule/pill shape */  
  
/* 4 corners alag – TL TR BR BL */  
.box { border-radius: 10px 20px 10px 20px; }  
  
/* Sirf top corners round karo */  
.card {  
    border-top-left-radius: 12px;  
    border-top-right-radius: 12px;  
    border-bottom-left-radius: 0;  
    border-bottom-right-radius: 0;  
}
```



// TOPIC 05

box-sizing Property

Content-box vs Border-box — sizing ka sahi tarika

KYA HAI?

box-sizing decide karti hai ki **width** aur **height** **kiska size define karein** — sirf content ka (default), ya content + padding + border sab ka milake. Yeh ek property bahut saari sizing problems solve kar deti hai!

content-box – Default ❌

width sirf **content** ka size.
Padding + border add hote hain
upar se — total size badh jaati hai!

200px + 40px pad + 10px border =
250px total 😱

border-box – Recommended ✔

width mein **content + padding + border** sab include. Jo size diya
wahi final size!

width: 200px = exactly 200px
total 🎯

box-sizing.css

```
/* — BEST PRACTICE — Har CSS file ki pehli lines — */
*,
*::before,
*::after {
    box-sizing: border-box; /* ← poori site ke elements */
}

/* content-box (default) — problem dikhata hai */
.box-a {
    box-sizing: content-box;
    width: 200px; padding: 20px; border: 5px solid blue;
    /* Actual = 200 + 40 + 10 = 250px */
}
```

```

/* border-box – size predictable! */
.box-b {
  box-sizing: border-box;
  width: 200px; padding: 20px; border: 5px solid blue;
  /* Actual = exactly 200px */
}

```

■ **Pro Tip:** `*, *::before, *::after { box-sizing: border-box; }` likhna ek **industry standard** hai — almost sab modern websites yahi karte hain. Hamesha CSS reset ke saath likho!

■ Outline vs Border — Difference

outline.css

```

.btn {
  border:      2px solid blue; /* space leta hai – layout affect */
  outline:    2px solid red;  /* space NAHI leta */
  outline-offset: 4px;        /* outline aur border ke beech gap */
}

/* Focus state – accessibility ke liye zaroori! */
.btn:focus {
  outline:      3px solid #06d6a0;
  outline-offset: 2px;
}

```

Feature	Border	Outline
Space leta hai	✓ Layout affect karta hai	✗ Space nahi leta
Box Model	✓ Part of box model	✗ Outside box model
Sides alag	✓ Har side alag style	✗ Sab sides ek saath

Feature	Border	Outline
<code>border-radius</code>	✓ Kaam karta hai	✗ Mostly nahi karta
Use Case	Design, card borders, dividers	Focus states, debugging

■ **Kabhi Mat Karo:** `outline: none` sirf tab hatao jab aapne khud custom focus style diya ho — warna keyboard users ke liye accessibility break ho jaati hai!

// QUICK REFERENCE

Box Model — Summary Table

Saare key points ek jagah

Property	Values / Example	Kya Karta Hai
padding	10px / 10px 20px / T R B L	Content ke andar ka space — background show karta hai
margin	10px / auto / -10px	Element ke bahar ka space — transparent
border	2px solid red	width + style + color — teeno zaroori hain
border-radius	8px / 50% / 999px	Rounded corners — 50% = circle
border-style	solid / dashed / dotted / double	Border line ka type
box-sizing	content-box / border-box	Width/height ka calculation — border-box use karo
margin: 0 auto	—	Element horizontally center karo (width zaroori)
outline	2px solid blue	Border jaisi lekin space nahi leta — focus states
outline-offset	4px	Outline aur element ke beech gap

 complete-real-world.css

```
/* — CSS Reset + Box Sizing — Hamesha Sabse Upar — */
*, *::before, *::after {
  margin: 0;
```



```
padding: 0;
box-sizing: border-box;
}

/* — Container — */
.container {
  max-width: 1200px;
  margin: 0 auto;      /* center */
  padding: 0 20px;     /* side breathing room */
}

/* — Card Component — */
.card {
  width: 320px;
  padding: 24px;
  margin: 16px;
  border: 1px solid #1e3a5f;
  border-radius: 12px;
  background: #0d1b2a;
}

/* — Button — */
.btn {
  padding: 10px 24px;
  border: 2px solid #e94560;
  border-radius: 6px;
  background: #e94560;
  color: white;
}
.btn:focus {
  outline: 3px solid #06d6a0;
  outline-offset: 2px;
}
```

► NEXT TOPICS TO COVER

CSS Selectors Deep Dive

Colors in CSS

Fonts & Typography

CSS Units (px, rem, em, %)

Display Property

Flexbox

CSS Grid

Positioning

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